**COMP2111 Database Design**
**Class Practice 6 Functional Dependencies**

Student ID: _____ Name: _____ Date: _____

The table shown lists dentist/patient appointment data.

- A patient is given an appointment at a specific time and date with a dentist located at a particular surgery.
- A patient can only have one appointment at a specific date.
- On each day of patient appointments, a dentist is allocated to a specific surgery for that day.

| *staffNo* | *dentistName* | *patNo* | *patName* | *appointment* | | *surgeryNo* |
|---|---|---|---|---|---|---|
| | | | | *Date* | *Time* | |
| S1011 | Tony Smith | P100 | Gillian White | 12/09/2026 | 10:00 | S15 |
| S1011 | Tony Smith | P105 | Jill Bell | 12/09/2026 | 12:00 | S15 |
| S1024 | Helen Pear | P108 | Ian MacKay | 12/09/2026 | 10:00 | S10 |
| S1024 | Helen Pear | P108 | Ian MacKay | 14/09/2026 | 14:00 | S10 |
| S1032 | Robin Plevin | P105 | Jill Bell | 14/09/2026 | 16:30 | S15 |
| S1032 | Robin Plevin | P110 | John Walker | 15/09/2026 | 18:30 | S13 |

State whether each of the following functional dependency is correct or not?

(1) staffNo, appointmentDate → surgeryNo (Yes)

Note that although (S1011, 12/09/2025), returns 2 occurrences of S15, this is only ONE value of S15, so, this is a valid functional dependency.

(2) staffNo, patNo, ~~surgeryNo~~ → dentistName, patName~~, appointmentDate, appointmentTime~~ (No)
For example, given (S1024, P108, S10), two appointment dates and time will be returned.

(3) patNo, appointmentDate → staffNo, dentistName, patName, appointmentTime, surgeryNo (Yes)

(4) staffNo, appointmentDate → dentistName, ~~patNo, patName, appointmentTime~~, surgeryNo (No)
For example, given (S1011, 12/9/2023), two appointment dates and time will be returned.

What is the candidate key for this relation?
(patNo, appointmentDate)

The table shown above is susceptible to update anomalies. Provide an example for each of insertion, deletion, and modification anomalies.

*Deletion Anomaly:*
*If we delete the details of the dentist "Helen Pear", we also lose the appointment details of the patient called "Ian MacKay".*

*Insertion anomalies:*

- *To insert a new patient that makes an appointment with the designated dentist, we need to enter the correct detail for the staff. For example, to insert the details of new patient in patientNo, patientName and an appointment, we must enter the correct details of the dentist (staffNo, dentistName).*
- *To enter new dentist data that doesn't have appointments to be assigned, we can't insert successfully because NULL values for the primary key is NOT allowed (entity integrity).*

*Modification anomalies:*
*With redundant data, when we want to change the value of one columns of a particular Dentist, for example the dentistName, we must update all the Dentist records that assigned to the particular patient otherwise the database will become inconsistent.*

*In fact, in the end, the design should like the following to avoid the anomalies:*

| *StaffNo* | *DentistName* |
|---|---|
| S1011 | Tony Smith |
| S1024 | Helen Pear |
| S1032 | Robin Plevin |

| *PatNo* | *PatName* |
|---|---|
| P100 | Gillian White |
| P105 | Jill Bell |
| P108 | Ian MacKay |
| P110 | John Walker |

| *StaffNo* | *PatNo* | *Appointment* | |
|---|---|---|---|
| | | *Date* | *Time* |
| S1011 | P100 | 12/09/2016 | 10:00 |
| S1011 | P105 | 12/09/2016 | 12:00 |
| S1024 | P108 | 12/09/2016 | 10:00 |
| S1024 | P108 | 14/09/2016 | 14:00 |
| S1032 | P105 | 14/09/2016 | 16:30 |
| S1032 | P110 | 15/09/2016 | 18:30 |
| S1032 | P110 | 12/09/2016 | 18:30 |

| *StaffNo* | *Appointment Date* | *SurgeryNo* |
|---|---|---|
| S1011 | 12/09/2016 | S15 |
| S1024 | 12/09/2016 | S10 |
| S1024 | 14/09/2016 | S10 |
| S1032 | 14/09/2016 | S15 |
| S1032 | 15/09/2016 | S13 |